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# Does ChatGPT provide better advice than robo-advisors?

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## Highlights

- ChatGPT provides better financial advice for one-time investments than robo-advisors.
- We compare the advice of ChatGPT and 17 robo-advisors to three investor profiles.
- ChatGPT's recommendations fit the three investor profiles and the benchmark.
- Only three robo-advisors almost meet the benchmark for all three investor profiles.
- Three robo-advisors miss the benchmark for all investor profiles.

## Abstract

We develop three investor profiles with different risk attitudes and consult ChatGPT and 17 robo-advisors to recommend an investment portfolio. We compare the recommendations with a benchmark derived from academic literature. ChatGPT's recommendations align with the three investor profiles and the benchmark. In contrast, only three out of the 17 robo-advisors come close to meeting the benchmark for all three investor profiles. Three robo-advisors fail to meet the benchmark for all investor profiles. Our findings reveal that ChatGPT provides better financial advice for one-time investments than robo-advisors. A policy implication is to clearly disclose that independent sophisticated chatbots can be recommended as a trustworthy source of information for retail investors.

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# Introduction

Identifying and using the most relevant information in financial decisions is something that many retail investors find challenging (Oehleretal., 2023). A significant percentage of retail investors suffer from information overload and insufficient financial literacy to effectively utilize the available information (Stolperand Walter,2017). As a result, these retail investors often rely on financial advisors (Foersteretal., 2017). Traditionally, financial advisors were human. However, a decade ago, robo-advisors were introduced to the market and have since become a permanent fixture, often as part of larger financial service providers such as banks, fund companies, and insurance companies. Robo-advisors were considered groundbreaking and promising, especially for their ability to provide financial advice with minimal behavioral biases typically associated with human advisors (D'Acunto etal., 2019; Foersteretal., 2017; Inderstand Ottaviani,2009; Linnainmaaetal., 2020). A robo-advisor typically constructs portfolios tailored to investors' preferences by asking questions about risk preferences, loss tolerance, investment amount, financial reserves, monthly disposable net income, and investment experience. Based on the answers to these questions, robo-advisors recommend an investment portfolio typically consisting of stock and bond index funds (D'Acunto and Rossi,2020; Hornetal., 2020; Hornand Oehler,2020; Kaya,2017; Oehleretal., 2022; Rossiand Utkus,2020).

Technological progress is opening new possibilities for financial advice. Companies in the financial service sector are expected to allocate more than half of their planned investment budget on artificial intelligence (AI) in 2023 and 2024, with a focus on improving customer-facing processes using chatbots (Sarathy,2023). Currently, the most discussed chatbot is ChatGPT. It was introduced in November 2020 and is a conversational variant of the third iteration of the Generative Pre-Trained Transformer (GPT) model (Brownetal., 2020). Kimetal.(2023) describe the model as pre-trained on a vast language corpus and then fine-tuned for specific tasks, such as summarization or sentiment analysis. GPT outperforms other existing models in summarization tasks (Bhaskaretal., 2022; Goyaetal., 2022) and performs impressively on financial literacy tests (Niszczotaand Abbas,2023). Therefore, it appears straightforward to analyze ChatGPT's ability as financial advisor. This has not been done yet although ChatGPT's utility in finance research and practical finance tasks have been the subject of other studies (e.g. Dowlingand Lucey 2023, Kim2023). We address this gap in the literature by analyzing the research question whether ChatGPT provides better advice than robo-advisors.

To answer this research question, we develop three investor profiles primarily differing in their risk attitudes. In April 2023, we utilized ChatGPT and 17 globally available robo-advisors to derive recommendations for each investor profile. We compare these recommendations with a benchmark from academic literature regarding a suitable portfolio structure.

Our findings reveal that ChatGPT provides better financial advice for one-time investments than robo-advisors. While only three out of the 17 robo-advisors come close to meeting the benchmark for all three investor profiles, three robo-advisors failed to meet the benchmark for any investor profile. ChatGPT, on the other hand, provides recommendations and additional information that aligned with the three investor profiles and the benchmark from academic literature. This is remarkable since ChatGPT used only a few targeted questions, and the consultation took significantly less time than answering the questions posed by the robo-advisors. Our findings are relevant not only to academics but also to regulators, investors, and providers of financial services.

We contribute to the literature on financial advice, AI for financial services, and financial information for retail investors in four ways. First, we derive three investor profiles from the academic and practitioner literature that can be used to analyze the suitability of financial advice. Second, we demonstrate the

economic usefulness of generative AI-based techniques for analyzing and summarizing investment information for different investor profiles in a meaningful way. Third, we provide preliminary evidence that chatbots can significantly improve the information provided to investors by enhancing presentation and content. ChatGPT retains and amplifies information content, allowing investors to focus on information that truly matters. Fourth, as investors often face significant information processing costs or investment commissions, the generation of investment approaches *via* ChatGPT could support investors' decision-making and increase investment returns.

Our findings imply that robo-advisors and even human advisors could use the information provided by ChatGPT to streamline their advice process and enhance the quality of advice. This finding can also be utilized by policies regarding retail investor information documents. Independent chatbots can be recommended as a trustworthy source for a second opinion to double-check recommendations provided by a human or robo-advisor. In summary, our findings indicate that generative AI systems have the potential to be the next meaningful advance in financial advice and investor information, which could be beneficial for service providers (lower costs) and investors (better advice).

Our study is organized as follows: In the next section, we review the related literature. In Section 3, we introduce the data and methodology to generate portfolio recommendations with ChatGPT and robo-advisors. Section 4 presents the empirical results. We discuss our findings and conclude in Section 5.

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## Section snippets

### ChatGPT and robo-advisors

ChatGPT employs AI to provide easily accessible information and has even been utilized in writing books about the usage of machine learning in finance, e.g., to develop trading strategies (Gautier, 2022; Osterrieder and ChatGPT, 2023). Furthermore, ChatGPT shares some advantages with robo-advisors, such as 24/7 availability and low consultation costs. These factors make both services intriguing to individuals with low net wealth who cannot afford a human advisor (D'Acunto and Rossi, 2020; ...

### Three investor profiles

In accordance with the literature (e.g. Oehler et al. 2019, Choi 2022, Foerster et al. 2017, Jacob et al. 2014) and tests and reviews of robo-advisors on websites and consumer protection organizations, we establish three investor profiles primarily differing in their risk attitude. These profiles need to be suitable for answering the questions posed by the robo-advisors and must have clear specifications (personal preferences) for ChatGPT. Simultaneously, the investor profiles should represent ...

### ChatGPT

The investment recommendations from ChatGPT for the three investor profiles are clear and align with the benchmark from the academic literature.

For investor profile A (high risk aversion), the portfolio recommendation is as follows: "60% in a bond fund (this would provide you with a reliable source of income while preserving your capital. Consider investing in a bond fund that holds high-quality bonds with low credit risk), 30% in an index fund (this would provide you with exposure to a...

## Discussion

ChatGPT provides investment recommendations that align with the consensus in the academic literature. As a robustness check, we conducted repeated queries to ChatGPT during our analysis to assess whether the recommendations remain consistent over time. We observed that the recommended asset allocations do change slightly over time (e.g., for investor profile C: 70% in equity funds, 15% in bond funds, and 15% in commodities and REITs). It's worth noting that these changes do not necessarily...

## Conclusion

Our analysis reveals the economic utility of generative AI-based techniques for analyzing and summarizing investment information for various investor profiles effectively. ChatGPT not only provides appropriate and well-founded portfolio recommendations for the three investor profiles but also delivers recommendations more quickly and with significantly less effort for the user. It is accompanied by easily understandable and valuable information about investing. An important policy implication...

## CRedit authorship contribution statement

**Andreas Oehler:** Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Matthias Horn:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing....

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper....

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